

REVIEW ARTICLE

Ancestry and Genetic Admixture of the Colombian Population: A Systematic Review and Meta-Analysis

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ABSTRACT

Objectives: A systematic review with meta-analysis was conducted to estimate and compare the proportions of Native American, African, and European ancestries in Colombia with those in other Latin American countries.

Materials and Methods: Articles published between 2005 and 2022 that estimated Colombian genetic admixture using autosomal markers (single nucleotide polymorphisms or ancestry informative markers) were reviewed. Meta-analyses were conducted at both national and regional levels. Finally, the results of Colombia were compared with estimates from other Latin American countries.

Results: Considering 75 records, Colombia displayed more balanced proportions of Native American and European ancestry (0.307 and 0.510, respectively), while the proportion of African ancestry was, on average, 2.23 times lower, with a value of 0.183. At the regional level, the distribution of these ancestral groups across the territory was not homogeneous. The Pacific region exhibits the highest African ancestry, followed closely by the Insular region. The Amazon region shows the highest Native American ancestry, followed by the Orinoquía region, while the Andean region presents the highest European ancestry, followed by the Caribbean and Orinoquía regions. In the Latin America context, Colombia ranks 8th for Native American ancestry, 5th for African ancestry, and 11th for European ancestry.

Conclusions: Colombia can be considered one of the most genetically diverse countries in Latin America, with significant variability in ancestry distribution both between and within regions. Further research is needed to fully understand Colombia's genomic ancestry and to develop precision public health strategies to advance precision medicine.

1 | Introduction

The year 1492 marks a pivotal moment in the history of the Americas, triggering one of the greatest transformations in human history: the mixing of multiple cultures (DANE 2007). The timing and extent of the intermixing among the diverse native groups inhabiting the Americas, the European colonizers from the Iberian Peninsula, and African slaves shaped the

different patterns of genetic diversity and ancestry in Latin American populations (Adhikari, Mendoza-Revilla, et al. 2016; Maldonado et al. 2023).

Colombia, as the gateway to South America, was one of the countries most influenced by this historical event (DANE 2007). Today, in the six Colombian regions (Insular, Caribbean, Pacific, Andes, Orinoquía, and Amazon),

populations of mixed ancestry, Afro-Colombians, the Romani or Gypsy people, and 115 native indigenous groups coexist (DANE 2022). Generally, the Andes and Caribbean regions concentrate most of Colombia's population, both before and after the European conquest, and are the country's most economically active regions. In contrast, the Pacific region and the forested area comprising the Orinoquía and Amazon regions are the largest in territory but the least populated and economically developed (De Castro and Restrepo 2015; Ossa et al. 2016). Among the indigenous groups, four tribes account for 58.1% of the indigenous population: *Wayuu*, *Zenú*, *Nasa*, and *Pastos*. The departments of Guajira, Córdoba, Sucre, Cauca, Nariño, and Chocó (Caribbean, Andes, and Pacific regions) hold half of the country's indigenous population, while the departments with the largest absolute populations are Guainía, Vaupés, Amazonas, Vichada, and Guajira (Amazon, Orinoquía, and Caribbean regions) (DANE 2019). The Afro-Colombian population can be categorized into four key groups: those located in the Colombian Pacific corridor (Pacific region), the Raizal communities of the Archipelago of San Andrés, Providencia, and Santa Catalina (Insular region), the community of San Basilio de Palenque (Caribbean region), and the population residing in municipal capitals or large cities (DANE 2007).

Several studies have genetically characterized Colombian populations and correlated the observed diversity with historical events (Ossa et al. 2016). Mitochondrial DNA and Y chromosome studies confirm the strong sexual bias during the early stages of admixture, predominantly involving European men from the Iberian Peninsula and Native or African women, reflecting the larger proportion of male European immigrants and their dominant social role (Bryc et al. 2010; Carvajal-Carmona et al. 2003, 2000; Homburger et al. 2015; Mathias et al. 2016). On the other hand, studies of autosomal markers, which provide more precise ancestry analyses, have shown a predominance of European ancestry linked to central/southern Spain (Chacón-Duque et al. 2018). However, the prevalence of continental ancestry components can vary significantly depending on the region within the country (Chande et al. 2020; Ossa et al. 2016; Ruiz-Linares et al. 2014; Salzano and Sans 2014). For example, genomic studies have shown that Medellín exhibits a higher percentage of European ancestry (62%), while Ibagué has a higher percentage of Native American ancestry (54%), and Chocó has a higher percentage of African ancestry (76%) (Chande et al. 2020; Conley et al. 2017; Criollo-Rayó et al. 2018; Medina-Rivas et al. 2016). Additionally, the subcontinental ancestry pattern can also vary within the country. In Chocó, the African ancestry is most closely related to populations from Nigeria, while Medellín shares ancestry with populations from Sierra Leone and Gambia (Conley et al. 2017). The Native American component of Colombians from the northwest is more closely related to the Chibcha-Páez groups, while the component of Colombians from the south is more closely related to the Andean groups from the Peruvian foothills (Chacón-Duque et al. 2018). Thus, the variability observed in the relative contribution of each ancestral population (continental and subcontinental) may result from different founding populations, different migration periods, geographical and cultural barriers to migration, the number of individuals involved in the

admixture process, and the duration of this process, among other factors (Chande et al. 2020; Moreno-Estrada et al. 2013).

Additionally, different patterns of genetic ancestry influence the variation in complex biomedical traits, highlighting the need to integrate genomic diversity into medicine (Belbin et al. 2018; Maldonado et al. 2023). For genomic medicine to reach its full potential, it is crucial that genomic studies are more representative of global populations. In this context, Colombia's genomic diversity represents a valuable and largely unexplored resource that could support the development of genomic medicine both locally and globally (Chande et al. 2021). Furthermore, the close relationship between ethnicity, ancestry, and health in Colombia reinforces the need to adopt the paradigm of precision public health as a means to implement genomic medicine in the country (Chande et al. 2021). Unlike precision medicine, which focuses on individual treatment, precision public health targets populations by relying on detailed genomic, demographic, and population profiles (Khouri et al. 2018, 2016). This approach allows for more effective resource allocation and facilitates the efficient implementation of emerging genomic technologies in these settings (Chande et al. 2021).

In this context, a systematic review with meta-analysis of studies on ancestry and genetic admixture in Colombia was conducted. The aim was to estimate the pooled proportions of continental ancestries (Native American, African, and European) and compare these findings with estimates from other Latin American countries using the same approach.

2 | Materials and Methods

2.1 | Literature Search, Inclusion/Exclusion Criteria, and Data Extraction

For the systematic review, studies using genomic data to assess the ancestry of Latin American populations were identified in the Scopus and Web of Science databases. The search equation, restricted to the fields of title, abstract, and keywords, was: (ancestry AND ("Latin American" OR "Latin America" OR "Latin Americans" AND genom*)). Duplicate records were removed, and studies that did not meet the selection criteria were excluded. Additionally, studies referenced in six reviews (Adhikari et al. 2017; Adhikari, Mendoza-Revilla, et al. 2016; De Oliveira et al. 2023; Rodrigues de Moura et al. 2015; Salzano and Sans 2014; Souza et al. 2019) and that met the selection criteria were included as records from other sources (Figure 1).

The inclusion criteria were: (i) studies using autosomal single nucleotide polymorphisms (SNPs) or autosomal ancestry informative markers (AIMs) to infer genetic ancestry; (ii) studies estimating Native American, African, and European genetic ancestry; (iii) studies reporting the number of individuals analyzed; and (iv) studies that provided the number of markers used. The exclusion criteria were: (i) studies using fewer than 10 SNPs or AIMs; (ii) studies involving related individuals; (iii) studies for which the full text was not available; (iv) review articles; (v) studies lacking data on the ancestry of the three ancestral populations (Native American, African, and European); and

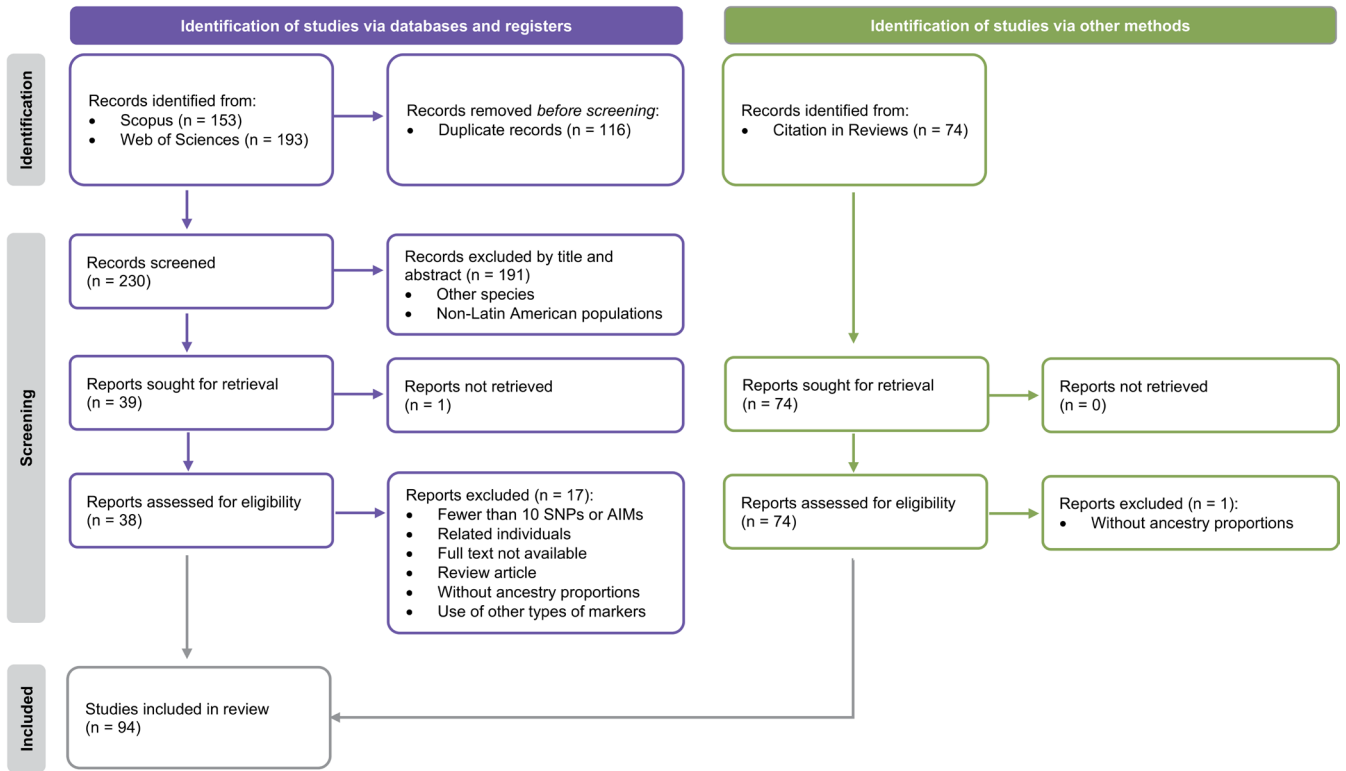


FIGURE 1 | PRISMA 2020 flow diagram for systematic review.

(vi) studies using only blood group markers, uniparental markers, or microsatellites (Figure 1).

From the selected studies, information related to the geographical location of the study population (country and sample location), sample size (N_s), markers used for ancestry inference (SNPs or AIMS), number of loci used, and percentage of continental ancestry (AFR, African ancestry; EUR, European ancestry; NAT, Native American ancestry) was extracted. Although the systematic review excluded studies that used only uniparental markers, some of the selected studies also included them. Therefore, to obtain a more comprehensive view of continental ancestry in Colombia, ancestry percentages for mitochondrial markers (mtDNA) and the Y chromosome were also extracted.

2.2 | Meta-Analysis

Since in some studies the sum of proportions is not equal to 1, mainly due to the inclusion of four continental ancestral populations, the values were standardized by dividing each contribution by the total sum of contributions (Rodrigues de Moura et al. 2015). Additionally, because some of the selected studies use the same cohort (e.g., 1000 Genomes Project, CANDELA Study, ChocoGen Project, EpiGen-Brazil Project, GALA Study, and HapMap Phase 3 Project), specific steps were taken to avoid counting individuals more than once in the meta-analysis. When multiple studies reported ancestry estimates based on identical samples (i.e., same cohort, with the same sample size and composition), the average of the reported ancestry proportions was calculated. In contrast, when studies used the same cohort but differed in sample size (indicating partial overlap or sample expansion), the ancestry

estimates from the study with the largest sample size were retained, thereby prioritizing representativeness and minimizing redundancy. Although we acknowledge that studies based on fewer genetic markers may produce lower-resolution estimates, the selection criterion was designed to maximize consistency in population-level representation. Notably, in six out of the 10 cases examined, the study with the largest sample size also employed the highest number of SNPs. Furthermore, even in cases where marker density varied, ancestry estimates were generally consistent. For example, in the Colombian sample of the CANDELA Study, Native American ancestry was estimated at 0.290 using 30 AIMS (Ruiz-Linares et al. 2014) and at 0.287 using 546,780 SNPs (Chacón-Duque et al. 2018), demonstrating minimal variation despite the increase in genotyping resolution.

To perform the meta-analysis and obtain unbiased estimates of ancestral proportions, ancestry estimates from each record were weighted according to the total census population of the sampling location. This adjustment accounts for the fact that the reviewed studies are based on samples collected at specific sites rather than through uniform sampling across each country, ensuring that the weighted ancestry proportions more accurately reflect population distributions (Adhikari et al. 2017; Eyheramendy et al. 2015). When a Latin American country or Colombian region had multiple records available, the methodology proposed by Eyheramendy et al. (2015) was applied, assigning to each record a weight (w_r) based on the ratio between the proportion of the total population (P_r) residing in sampling location (according to census data) and the proportion of the total sample (S_r) contributed by that record:

$$w_r = \frac{P_r}{S_r}$$

For example, the record corresponding to the Northeastern region of Argentina, reported by Corach et al. (2010), includes four provinces and represents 0.0377 of the total population, while contributing 0.0373 of the total sample. Thus, the weight for this record is calculated as (Table S1):

$$w = \frac{0.0377}{0.0373} = 1.01$$

A weight greater than 1 indicates that a sampling location is underrepresented in the sample relative to its population size, and its contribution is therefore amplified. Conversely, a weight less than 1 reflects overrepresentation, and the record is down-weighted accordingly. Once record weights are defined, ancestry proportions for each Latin American country or Colombian region are computed as weighted averages of ancestry estimates:

$$\bar{p}_{\text{ancestry}} = \frac{\sum_{r=1}^n w_r \times p_r}{\sum_{r=1}^n w_r}$$

where p_r is the ancestry proportion (e.g., Native American, European, or African) of each record, and w_r is the weight assigned to each record.

Finally, maps and triangular plots of the estimated proportions of Native American, African, and European genetic ancestry were generated using the R packages *ggplot2* (Wickham 2016) and *ggtern* (Hamilton and Ferry 2018), respectively.

3 | Results

3.1 | Characteristics of the Selected Studies

After conducting a global literature search that covered articles published between 2005 and 2022, 94 studies were selected and examined for a comprehensive review. These studies assessed continental genetic ancestry (Native American, African, and European) based on autosomal markers in 16 Latin American countries. Brazil had the highest number of records (84), followed by Colombia (75) and Mexico (28). The countries with the fewest records were Honduras, Paraguay, the Dominican Republic, and Uruguay, each with only one record. Additionally, 63 studies used AIMs, while 31 used SNPs. The number of autosomal markers varied widely across studies, particularly for SNPs. In studies using AIMs, the number of loci ranged from 11 to 1814, whereas in studies using SNPs, it ranged from 100 to 700,093. Finally, eight of the selected studies also estimated continental genetic ancestry using uniparental markers (mtDNA and the Y chromosome) for Colombia. Tables S1–S3 provide a summary of all the studies included in the meta-analysis that assessed continental genetic ancestry based on autosomal, mtDNA, and Y chromosome markers, respectively.

3.2 | Ancestry and Genetic Admixture in Colombia

The weighted proportions of Native American, African, and European ancestry based on autosomal, mtDNA, and Y chromosome markers are summarized in Figure 2. For

autosomal markers, the proportions of European (0.510) and Native American ancestry (0.307) were more similar, whereas the proportion of African ancestry was on average 2.23 times lower, with a value of 0.183. In contrast, uniparental marker results reveal an imbalance between maternal and paternal ancestry, with Native American mitochondrial haplogroups (0.870) and European Y chromosome haplogroups (0.770) being the most predominant (Figure 2).

Genetic ancestry analyses across Colombian geographic regions covered 23 departments within six regions, with the Andean and Pacific regions having the highest number of records (44 and 9, respectively, as shown in Table 1). The Amazon region exhibited a higher prevalence of Native American ancestry (0.682), while the Pacific and Insular regions showed a higher prevalence of African ancestry (0.668 and 0.627, respectively). In contrast, the Andean and Caribbean regions showed a greater proportion of European ancestry (0.513 and 0.455, respectively), and the Orinoquía region stood out for having similar contributions from Native American and European ancestries (0.469 and 0.444, respectively; Table 1 and Figure 3A–C). Additionally, the triangular plot revealed significant variability within the Andean, Caribbean, and Pacific regions. In the Andean and Caribbean regions, records range from those closest to the European vertex to those nearest the Native American vertex. Meanwhile, in the Pacific region, records vary from those closest to the African vertex to those with similar contributions of Native American and European ancestry (Figure 3D).

3.3 | Ancestry and Genetic Admixture in Latin America

The weighted proportions of the three continental ancestries in Latin American countries, as estimated by the meta-analysis, are summarized in Table 2. To better represent genetic admixture in Latin America, the table also includes admixture proportions for other countries (Honduras, Paraguay, Dominican Republic, and Uruguay) that have only one study.

Mexico, the only North American country included in Latin America, exhibited a higher prevalence of Native American ancestry (0.599). Among the five Central American countries analyzed, three showed a greater European contribution (Cuba: 0.736, Costa Rica: 0.702, and Puerto Rico: 0.676), one had a predominantly African contribution (Honduras: 0.810), and one displayed more balanced proportions of European and African ancestry (Dominican Republic: 0.520 and 0.380, respectively). In contrast, South American countries exhibited greater variability. Three countries had a higher contribution of European ancestry (Brazil: 0.753, Argentina: 0.730, and Uruguay: 0.687), while three others had a greater prevalence of Native American ancestry (Bolivia: 0.910, Peru: 0.766, and Ecuador: 0.615). Additionally, four countries displayed more comparable proportions of European and Native American ancestry (Chile: 0.527 and 0.439, respectively; Venezuela: 0.515 and 0.380, respectively; Colombia: 0.510 and 0.307, respectively; and Paraguay: 0.554 and 0.338, respectively) (Table 2).

On the other hand, both the map and the triangular plot show that, overall, the Latin American countries with the highest

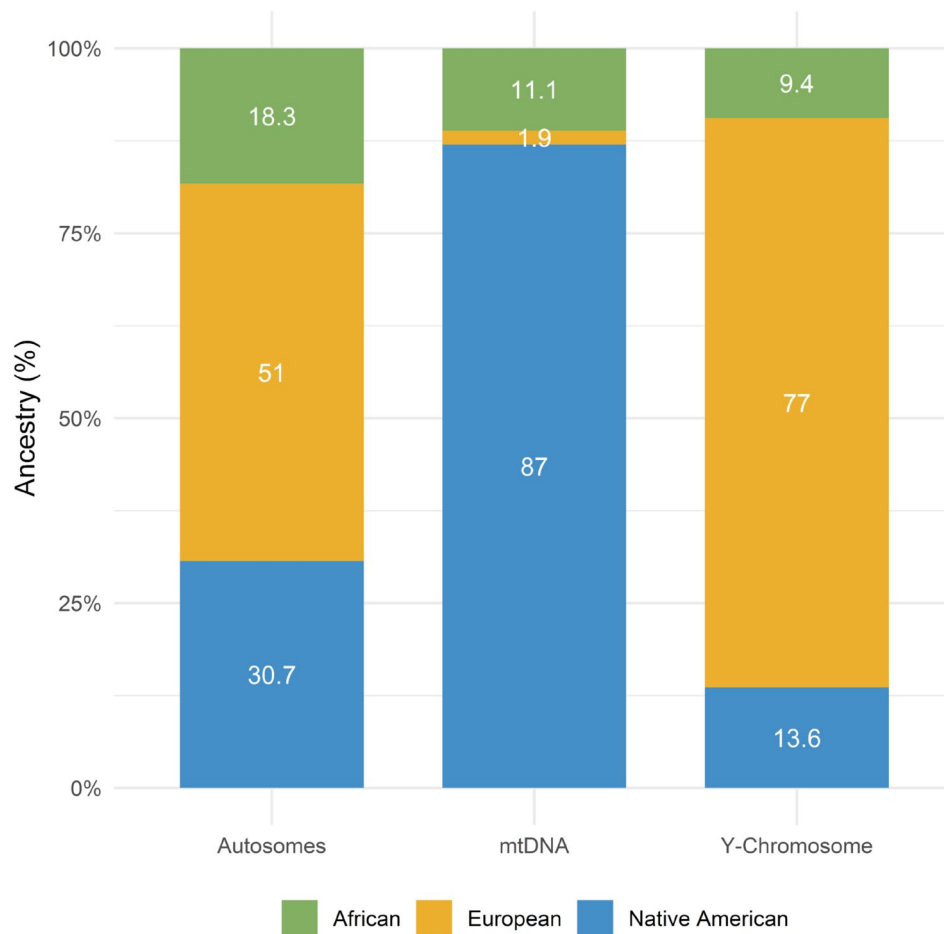


FIGURE 2 | Continental ancestry proportions for Colombia obtained through meta-analysis and estimated using autosomal, mtDNA, and Y chromosome markers.

TABLE 1 | Continental ancestry proportions obtained through meta-analysis by region of Colombia, based on studies using autosomal genetic markers.

Region	Number of records	Ancestry proportions		
		NAT	AFR	EUR
Amazon	5	0.682	0.070	0.248
Andean	44	0.376	0.111	0.513
Caribbean	6	0.274	0.270	0.455
Insular	1 ^a	0.113	0.627	0.260
Orinoquía	3	0.469	0.087	0.444
Pacific	9	0.156	0.668	0.176

Abbreviations: AFR, African ancestry; EUR, European ancestry; NAT, Native American ancestry.

^aProportion derived from the only study conducted for the region.

Native American ancestry are Bolivia (0.910) and Peru (0.766), while Honduras exhibits the highest African ancestry (0.810). In contrast, Brazil, Cuba, and Argentina have the highest European ancestry (0.753, 0.736, and 0.730, respectively, Figure 4A–D).

4 | Discussion

Although several studies have demonstrated that the proportion of genetic admixture from the three continental ancestral populations varies widely among Latin American countries and have even highlighted genetic ancestry differences between regions within each country (e.g., Adhikari, Mendoza-Revilla, et al. 2016; Bryc et al. 2010; De Oliveira et al. 2023; Homburger et al. 2015; Rodrigues de Moura et al. 2015; Ruiz-Linares et al. 2014; Salzano and Sans 2014), few efforts have consolidated and updated this information through a region-specific meta-analysis focused on Colombia. Therefore, given the increasing use of human genomic data in recent decades, this study aimed to perform a comprehensive literature review focusing on research employing autosomal SNPs or AIMs to infer genetic ancestry, disaggregated by geographic region within Colombia. This aims to enhance the understanding of the genetic heterogeneity across different geographic regions of Colombia and to compare these findings with estimates from other Latin American countries. However, it is important to acknowledge that genetic ancestry estimates from the studies included in this meta-analysis often rely on convenience samples that are not necessarily representative of entire populations, which may introduce biases due to limited geographic and socioeconomic coverage (Adhikari et al. 2017).

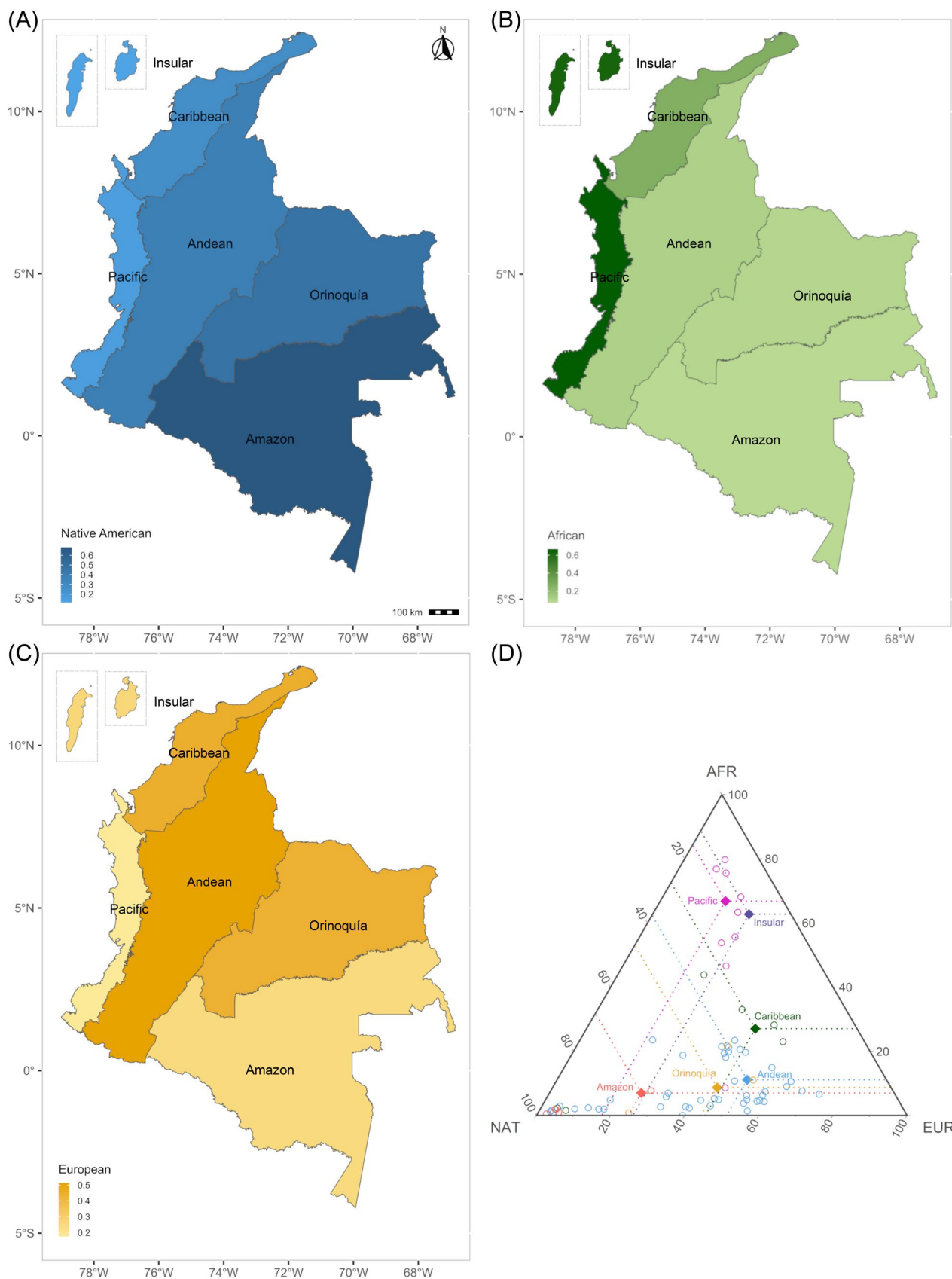


FIGURE 3 | Continental ancestry proportions obtained through meta-analysis by region of Colombia, based on studies using autosomal genetic markers. (A) Native American ancestry (NAT); (B) African ancestry (AFR); (C) European ancestry (EUR); and (D) triangular plot illustrating the clustering of genetic ancestry. Circles represent estimates from individual studies, while diamonds indicate estimates derived from the meta-analysis. Records from the Caribbean region are shown in dark green, Andean in light blue, Pacific in bright pink, Orinoquia in bright orange, Amazon in soft red, and Insular in dark blue.

TABLE 2 | Continental ancestry proportions obtained through meta-analysis for Latin American countries, based on studies using autosomal genetic markers.

Country	Number of records	Ancestry proportions		
		NAT	AFR	EUR
Argentina (ARG)	8	0.231	0.038	0.730
Bolivia (BOL)	9	0.910	0.005	0.085
Brazil (BRA)	84	0.087	0.159	0.753
Chile (CHL)	9	0.439	0.035	0.527
Colombia (COL)	75	0.307	0.183	0.510
Costa Rica (CRI)	3	0.275	0.023	0.702
Cuba (CUB)	8	0.032	0.232	0.736
Dominican Republic (DOM)	1 ^a	0.090	0.380	0.520
Ecuador (ECU)	5	0.615	0.076	0.309
Honduras (HND)	1 ^a	0.170	0.810	0.020
Mexico (MEX)	28	0.599	0.035	0.366
Paraguay (PRY)	1 ^a	0.338	0.108	0.554
Peru (PER)	4	0.766	0.028	0.206
Puerto Rico (PRI)	10	0.131	0.192	0.676
Uruguay (URY)	1 ^{as}	0.222	0.091	0.687
Venezuela (VEN)	5	0.380	0.105	0.515

Abbreviations: AFR, African ancestry; EUR, European ancestry; NAT, Native American ancestry.

^aProportions derived from the single study conducted for the country.

Considering all records from Colombia, the weighted proportions of Native American, African, and European ancestry based on autosomal markers were 0.307, 0.183, and 0.510, respectively. Most previous studies analyzing individuals from different Colombian regions have reported higher proportions of European ancestry (0.580–0.625) compared to Native American (0.274–0.360) and African ancestry (0.090–0.110) (Adhikari, Mendoza-Revilla, et al. 2016; Bryc et al. 2010; Chacón-Duque et al. 2018; Homburger et al. 2015; Ruiz-Linares et al. 2014). Only one study found a higher proportion of Native American ancestry than European ancestry (0.506 and 0.332, respectively) (Silgado-Guzmán et al. 2022). However, the results of this study align with the estimates from the meta-analysis by Rodrigues de Moura et al. (2015), which describe more balanced proportions of European and African ancestry in Colombians (0.440 and 0.420, respectively). Thus, Colombia can be considered one of the most genetically heterogeneous Latin American countries, where populations with multiple ancestries coexist, including Afro-descendants and 115 indigenous native groups (DANE 2022).

In the Latin American context, Colombia ranks eighth for Native American ancestry, fifth in African ancestry, and 11th in European ancestry. The highest proportions of Native American ancestry were found in the Central Andes (Peru, Bolivia, and Ecuador) and in Mexico, while the highest African ancestry contributions

were observed in Central America (Honduras), the Caribbean (Dominican Republic, Cuba, and Puerto Rico), and Brazil. European ancestry predominated in southern South America (Brazil, Argentina, and Uruguay) and the Caribbean (Cuba and Puerto Rico). These findings are consistent with previous reviews (Adhikari et al. 2017; De Oliveira et al. 2023; Rodrigues de Moura et al. 2015; Salzano and Sans 2014). While the high observed heterogeneity may largely stem from differences in sample sizes and study design (Rodrigues de Moura et al. 2015; Souza et al. 2019), it also reflects the complex history of colonization in each country, which led to different admixture processes.

Following the arrival of Europeans, it is estimated that approximately 90% of the Native American population collapsed. This led to the annihilation of native groups in areas with relatively small populations (the Caribbean and southern South America) and the settlement of Europeans in areas densely populated by natives to exploit them as labor (Mesoamerica and the Central Andes) (Adhikari et al. 2017; Salzano and Sans 2014). Since the major centers of European colonial power were located within the territories of the Inca (Peru, Bolivia, and Ecuador) and Aztec (Mexico) empires (Adhikari et al. 2017; Rodrigues de Moura et al. 2015), it is likely that the natives in these regions were more extensively involved in the admixture process than other indigenous populations. Thus, this historical context could have contributed to the high proportions of Native American ancestry observed in these countries today. Additionally, north of the Andes, the Chibchan civilization thrived and was regarded as the third most developed group in the Americas after the Aztecs and Incas (Ossa et al. 2016). Consequently, it is to be expected that the Colombian population would also exhibit a substantial proportion of Native American ancestry. However, the admixture process in Latin America was characterized by strong sexual bias, with interethnic unions taking place primarily between European colonizers and Native American women. This pattern has been widely documented in various studies (Acosta-Uribe et al. 2022; Carvajal-Carmona et al. 2000; Córdoba et al. 2012; Kehdy et al. 2015; Rojas et al. 2010; Xavier et al. 2015) and is further supported by meta-analyses of uniparental markers, which reveal a higher proportion of Native American ancestry in maternal lineages and greater European ancestry in paternal lineages.

On the other hand, with the collapse of native populations, the introduction of African slaves was promoted to exploit the lands taken from the natives and to work in large-scale colonial agricultural production, primarily sugarcane (Adhikari et al. 2017; Rodrigues de Moura et al. 2015). Estimates suggest that approximately 42% of the slaves arrived in Brazil, around 25% in British colonies, and about 15% in Spanish colonies (Adhikari et al. 2017). Consequently, these slaves played a crucial role in the historical and genetic composition of Caribbean countries, Central America, and Brazil (Rodrigues de Moura et al. 2015; Salzano and Sans 2014), as evidenced by the high proportions of African ancestry in these countries.

Although the overall analysis displayed more balanced proportions of European and Native American ancestry in Colombia, the distribution of these ancestral groups across the territory was not homogeneous. Various studies and reviews have highlighted the high variability in continental ancestry among Colombian populations and regions without reaching

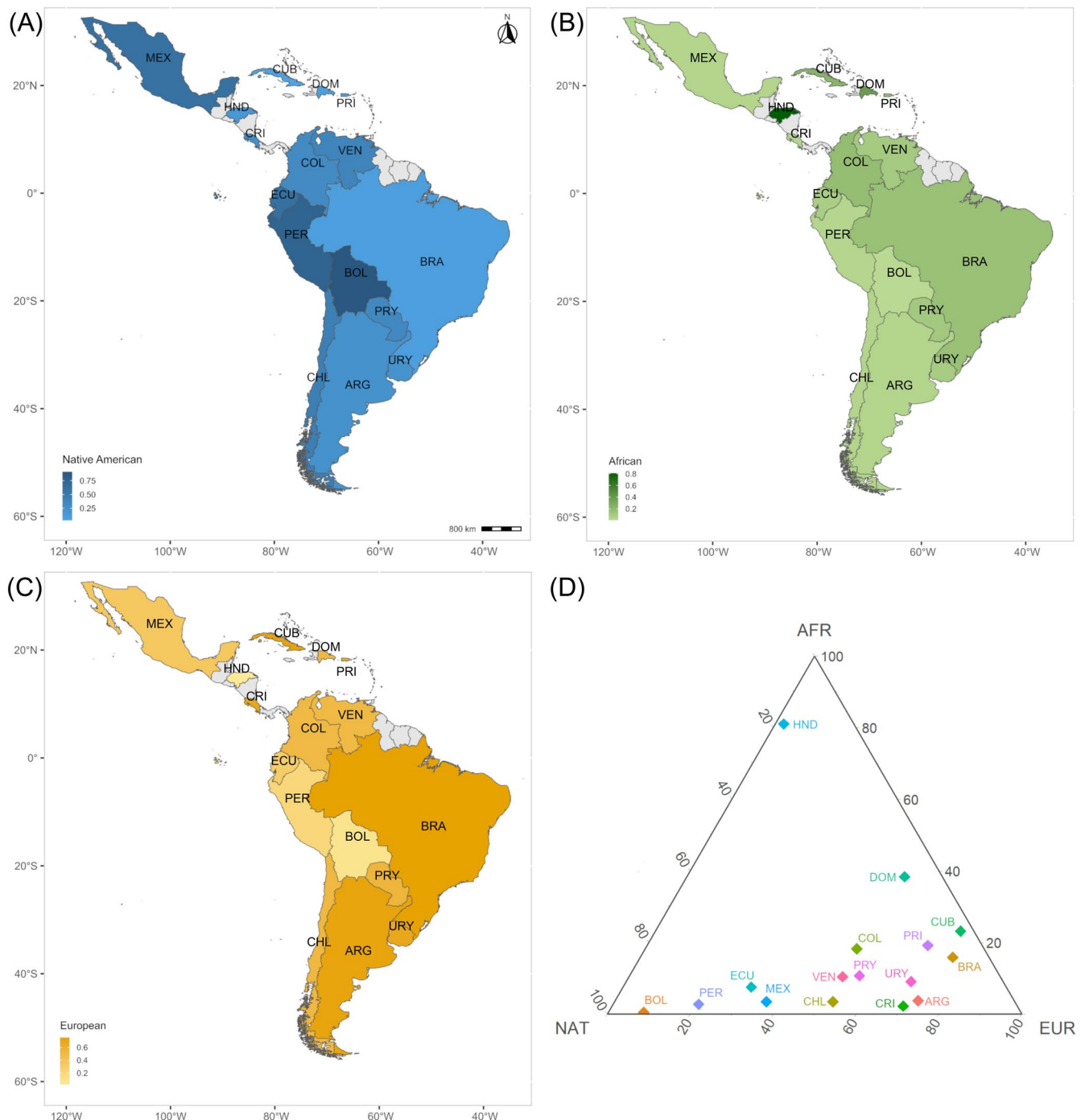


FIGURE 4 | Continental ancestry proportions obtained through meta-analysis for Latin America, based on studies using autosomal genetic markers. (A) Native American ancestry (NAT); (B) African ancestry (AFR); (C) European ancestry (EUR); and (D) triangular plot illustrating the clustering of genetic ancestry. Diamonds represent the estimates obtained in the meta-analysis. Country abbreviations are according to Table 2.

definitive conclusions (Bryc et al. 2010; De Oliveira et al. 2023; Ruiz-Linares et al. 2014; Salzano and Sans 2014). However, in this meta-analysis, considering each ancestral component, the Pacific region of Colombia exhibits the highest African ancestry, followed closely by the Insular region. The Amazon region shows the greatest Native American ancestry, followed by the Orinoquía region, while the Andean region presents the highest European ancestry, followed by the Caribbean and Orinoquía regions. These findings are consistent with the estimates reported by Ossa et al. (2016), who, using 46 AIMs and 761 unrelated individuals from five of the six regions, found

that the Pacific had the highest African ancestry, the Amazon had the highest Native American ancestry, and the Andean, Caribbean, and Orinoquía regions had the greatest proportion of European ancestry. Similarly, the results of this study align with the estimates reported by Ruiz-Linares et al. (2014). These researchers, using 30 AIMs and 1713 unrelated individuals from different regions, found that coastal regions (particularly the Pacific) have the highest African ancestry, central areas have the highest European ancestry, and the southwest and east of the country (Amazon) have the highest Native American ancestry.

The notable heterogeneity observed among Colombian regions may be explained by their distinct colonization histories, which led to different admixture processes. For instance, the high European ancestry in Andean populations aligns with the fact that during European colonization, the Magdalena River, located between the Central and Western branches of the Andes, was the primary route for expansion into the interior of the country (Ossa et al. 2016). Additionally, geographically isolated areas with low population densities likely received little non-native immigration, resulting in higher proportions of Native American ancestry, as seen in the Amazon region (Ossa et al. 2016; Ruiz-Linares et al. 2014). In contrast, the islands of the San Andrés and Providencia archipelago in the Insular region, which originally served as commercial and military bases, evolved into plantations sustained by enslaved African labor from Jamaica and other Caribbean islands (Eastman 1922). Given that the enslaved population constituted more than 50% of the total population at the beginning of the 19th century (Eastman 1922), it is expected that the Raizal population in this region shows high proportions of African ancestry. Similarly, the considerable proportion of African ancestry in the Pacific region corresponds with historical records indicating that during the 16th and 17th centuries, a substantial number of slaves were brought to this region to work in gold mines, leading to a predominantly African population during that period (Ossa et al. 2016).

Additionally, the results of this analysis highlight significant variability in ancestry within the Pacific, Caribbean, and Andean regions. In the Pacific, some populations exhibit high African ancestry, while others show a more balanced contribution of Native American and European ancestry. This finding is consistent with data from the 2018 National Quality of Life Survey, which shows that this region has the highest proportions of individuals who self-identify as Indigenous or as Black, Afro-Colombian, Raizal, or Palenquero in Colombia (DANE 2021, 2022). In contrast, the Andean and Caribbean regions display a different pattern of ancestry diversity, with certain populations showing high European ancestry, while others exhibit significant Native American ancestry. This pattern in the Andean Region was previously observed by Ossa et al. (2016), who found that the Central West subregion of the Andes had the highest contributions of European ancestry, whereas the Southeast and Southwest subregions retained significant Native American ancestry. Additionally, it is important to mention that within Colombia, there is also a notable geographical differentiation in subcontinental Native American ancestry (Chacón-Duque et al. 2018; Mogollón Olivares et al. 2020). Specifically, in the northwest of Colombia, Native American ancestry is more closely related to the Chibchan-Páez natives, whereas in the south, it is more similar to the Peruvian Andean Piedmont component (Chacón-Duque et al. 2018).

Considering the geographical distribution of the studied populations, there are few records for the Insular, Amazon, Orinoquía, and Caribbean regions of Colombia. Therefore, additional studies with larger sample sizes from these areas are needed to better understand and describe the genetic ancestry of the country's different regions.

The extensive and variable admixture observed in Latin American populations not only underlies extensive genetic

heterogeneity but also contributes to high phenotypic diversity, which may influence disease risk patterns in the region (Adhikari, Fontanil, et al. 2016). While genomic diversity studies and their effects on phenotypes have predominantly focused on European populations (Maldonado et al. 2023; Popejoy and Fullerton 2016), various genetic studies have demonstrated that the risk of certain diseases among Latin Americans varies according to continental ancestry (Goetz et al. 2014; Soares-Souza et al. 2018).

For example, several associations between complex diseases and continental ancestry have been observed: (i) In Latin America, higher Native American ancestry in the 6q25 and 11p15 regions is associated with a reduced risk of non-familial breast cancer (Fejerman et al. 2012). (ii) In Mexicans and other Latin American countries, Native American ancestry is associated with lower odds of asthma, while African ancestry is associated with higher odds (Pino-Yanes et al. 2015). (iii) In Mexico, Native American ancestry is linked to a higher prevalence of type 2 diabetes (Martinez-Marignac et al. 2007). (iv) In Costa Rica, African ancestry was associated with an increased risk of myocardial infarction and hypertension, whereas Native American ancestry was associated with an increased risk of metabolic syndrome and fasting glucose abnormalities (Ruiz-Narváez et al. 2010). (v) In Brazil, Native American ancestry was correlated with the risk of developing B-cell acute lymphoblastic leukemia, while European ancestry had a protective effect against this childhood leukemia (Carvalho et al. 2015). (vi) In Argentina, Native American ancestry was associated with a substantial increase in susceptibility to systemic lupus erythematosus (Seldin et al. 2008).

For Colombia, specifically in Antioquia and Chocó, a significantly positive correlation has been observed between European ancestry and estimates of the polygenic risk score for gastric cancer; between African ancestry and the predicted risk for type 2 diabetes, colorectal cancer, and prostate cancer; and between Native American ancestry and the predicted risk for acute lymphoblastic leukemia (Chande et al. 2021, 2017). Additionally, in the Andean region, a statistically significant association has been observed between Native American ancestry and the expression of the *ERBB2* gene at locus 17q12. This indicates that women with higher Native American ancestry are more likely to develop luminal breast cancer with higher *ERBB2* expression compared to women with higher European ancestry (Serrano-Gómez et al. 2017). In Medellín, several genes related to the innate and adaptive immune systems have been found in chromosomal regions enriched with African ancestry (e.g., *HLA-A*, *HLA-DRB51*, *HLA-DRB5*, *HLA-B*, and *MICA*) and Native American ancestry (e.g., *CD226* and *MAPK10*), which participate in the defense against endemic pathogens such as malaria and other agents (Norris et al. 2020; Rishishwar et al. 2015). Similarly, in Santander, Native American ancestry showed a protective association against Chagas disease within the major histocompatibility complex region, suggesting that in ancestral populations with higher proportions of Native American ancestry, long-term exposure may have provided a more efficient immune response against the parasite (Casares-Marfil et al. 2021). Additionally, in different regions of Colombia, 21 pathogenic variants of multiple ancestral origins have been identified in genes associated with Alzheimer's disease and frontotemporal

lobar degeneration, with the *PSEN1* gene harboring many of these variants (Acosta-Uribe et al. 2022).

Finally, given the strong association between ancestry and disease risk observed across different Latin American countries, it is crucial to advance genomic research in populations with high genetic heterogeneity, such as those in Colombia. This will support the development of precision public health strategies to advance in the field of precision medicine.

5 | Conclusions

The compilation of various studies on genetic ancestry of Colombian populations using a meta-analytic approach allows us to conclude that Colombia can be considered one of the most genetically diverse countries in Latin America. Although the weighted proportions of Native American, African, and European ancestry in the general population were 0.207, 0.183, and 0.510 respectively, the distribution of these ancestral groups across the territory was not homogeneous. Thus, we highlight that there is high variability in ancestry both between and within different regions of the country. However, more studies are needed to understand the complete genomic ancestry landscape of Colombia and to develop precision public health strategies that advance the field of precision medicine.

Author Contributions

Ghennie T. Rodríguez-Rey: conceptualization (lead), data curation (lead), formal analysis (lead), investigation (equal), methodology (equal), writing – original draft (lead), writing – review and editing (lead). **Marcela Orjuela-Rodríguez:** conceptualization (supporting), data curation (supporting), investigation (equal), methodology (equal), writing – review and editing (supporting). **Narmer Fernando Galeano-Vanegas:** conceptualization (supporting), funding acquisition (lead), investigation (equal), methodology (equal), writing – review and editing (supporting).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article, as no new datasets were generated. However, the information collected for the meta-analysis and supporting the findings of this study is available in the [Supporting Information](#) of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Continental ancestry proportions based on autosomal genetic markers in Latin American countries. **Table S2:** Continental ancestry proportions based on mtDNA in Colombia. **Table S3:** Continental ancestry proportions based on Y-Chromosome in Colombia.